

Technical Proposal - NSO 250 MW / 1000 MWh Standalone BESS

DRAFT PROPOSAL PREPARED BY THE BIDDER'S ADVISOR - NOT AN ENVISION DOCUMENT AND NOT AN OFFER. This draft assembles, from the material Envision has supplied to date, the technical proposal that most closely fits the tender. Text in BLACK is sourced from a received Envision or NSO document. Text in RED is DRAFTED GAP-FILL: it was NOT found in any received Envision document and is written the way the tender requires it to read, so that Envision can confirm it, correct it, or replace it. No red statement has been verified with Envision and none may be relied upon or submitted until Envision has adopted it in writing.

SOURCE DOCUMENTS GOVERN. Where this draft and a tender document, an Envision document or an issued addendum or clarification disagree, that document governs and this draft is wrong. Clause references are to NSO RFP Volumes I-III, Addendum No. 01 (7 August 2026), Annex A Functional and Performance Requirement, and the 76-item clarification register (21 August 2026), all held in the corpus with SHA-256 recorded in MANIFEST.sha256.

0. Document control

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Document control

Control field	Controlled value
Tender	TR/REP&PM/ICB/2026/001/C
Tender title	Establishment of 250 MW / 1000 MWh Standalone Battery Energy Storage System from 10 MW / 40 MWh AC Capacity Projects on Build, Own and Operate (BOO) Basis with 15 Year Operational Period
Document	Technical Proposal - Volume II Section 6 response and supporting schedules
Offered by	Envision Energy / the supplying group entity (DRAFT - not yet adopted)
Project Proponent	Hayleys
Submission deadline	4 September 2026, 10.00 hrs (Addendum No. 01 item 01)
Clarification window	CLOSED 25 August 2026 - no further questions to NSO are possible
Declared Capacity offered	10 MW / 40 MWh, export-limited (see section 2)
Installed configuration	11 MW / 44 MWh hardware - intentional oversizing
Status	DRAFT FOR ENVISION COMPLETION - not an Envision document

Control field	Controlled value
Completion required	Every red entry requires Envision confirmation before submission

1. Offer summary and the single design decision behind it

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Envision offers the 11 MW / 44 MWh equipment configuration, declared to NSO as a 10 MW / 40 MWh Contracted Capacity and held to 10 MW by a tested export limiter. This is deliberate oversizing, and it is the single decision that resolves most of the tender's binding constraints at once.

Tender constraint	10 MW / 40 MWh build	11 MW / 44 MWh build, declared at 10 MW
Usable energy at BoL against 40 MWh required	40.2 MWh - 0.5 % margin, smaller than the design calculation's own stated 1-2 % variability	43.9 MWh - approx. 9.75 % margin
Round-trip efficiency, year 15, auxiliary-inclusive (Volume I §2.8(iii), floor 85 %)	85.0 % - exactly on the floor, no margin	85.0 % declared; re-rated at the lower relative loading of the oversized build, target 85.6 % - Envision to confirm
Reactive capability (clarification 24: +/-0.3 x rated active power)	+/-3.29 Mvar against +/-3.0 Mvar required - passes	+/-3.29 Mvar against +/-3.0 Mvar required at a declared 10 MW - passes with margin
Four hours at rated output at site ambient (Volume I §3.1(k), envelope to +45 C)	Marginal - the 4.02 h figure is computed at 35 C	Held at +45 C on the oversized energy base - Envision to confirm by thermal calculation, see section 5
Minimum Dispatchable Storage Capacity, year 15 (68 % of contracted)	30.8 MWh against 27.2 MWh required	Approx. 33.6 MWh against 27.2 MWh required - Envision to confirm the degradation curve on the 44 MWh base
400 full equivalent cycles per year, incl. ancillary-service energy (clarification 55(b))	Duty falls on a thinner energy base, so cell throughput per contracted MWh is higher	Lower depth of discharge for the same delivered energy, extending cycle life - the principal answer to the 45 C exposure at section 5

Source: Envision design calculations of 29 July 2026 (10 MW / 40 MWh, superseded 5 August) and 5 August 2026 (11 MW / 44 MWh); NSO RFP Volume I; Annex A; clarification register items 24 and 55.

2. Declared Capacity, export limiting and the second technical proposal

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Clarification 2 confirms that the Contracted Capacity in Volume III Appendix 01 Clause 1 will be the Declared Capacity offered in Section 1 of Volume II,

amended into the Contract at execution. Clarification 66 confirms the declared capacity must be achieved and demonstrated at commissioning.

1. Declared Capacity: 10 MW / 40 MWh. Every contractual obligation - availability, round-trip efficiency, dispatchable capacity, cycle allowance - is assessed against this figure.
2. Installed equipment: 11 MW / 44 MWh, comprising 4 x ENPCS2520 on one AC skid (10.08 MW) plus the DC container set described in section 3.
3. Export limiter: the EnOS PPC limits net active power at the revenue metering point to 10.0 MW continuous, with a hard ceiling of 11.0 MW (10 MW +10 %) at the Termination Point. Setpoint resolution 0.1 MW, response time under 1 s to a step, fail-safe action to 0 MW on loss of measurement or loss of PPC heartbeat. Redundant measurement is taken from the revenue metering CT/VT set. Envision to confirm the limiter hierarchy, response time and fail-safe state.
4. Second technical proposal: Addendum No. 01 item 23 permits a maximum of two Technical Proposals, and clarification 73(a) confirms up to two alternative complete solutions with no interchange of major equipment between them. Bidder decision pending Envision's answer on the grid-forming EMT model (section 7): if a grid-forming PSCAD model of the ENPCS2520 does not exist, the second proposal should carry a converter for which one does.

3. Equipment schedule and configuration

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All part numbers, ratings and dimensions below are as recorded in the Envision design calculations, the filled Volume 2 Guaranteed Technical Particulars and the 21 August evidence dossier.

Item	Designation	Rating and particulars
Battery		
Cell	HC-L755A, LFP	3.2 V / 755 Ah / 2416 Wh. Approx. 10,000 cycles to 70 % retention at 25 C, 0.25P. Storage recovery 99 % at 25 C and 97 % at 45 C after 135 days at 100 % SOC.
Module	ENS-1P416S-L-10	Rack configuration 416S1P x 6.
DC container (primary)	ENS-D06G-24120-10100-000	6,030 kWh per container; 6 racks in parallel; 24.121 MWh nameplate per AC twin-skid; 4 containers per twin-skid. DC range 1165-1500 V.
DC container (secondary)	ENS-D10G-20100-10100-000	Supersedes ENS-D10E. 10 racks per DC container. DC range 1165-1500 V.
Container envelope	-	6058 (L) x 2438 (W) x 3258 (H) mm, under 50,000 kg.
Power conversion		
PCS	ENPCS2520	2.52 MVA, 690 V, 50 Hz, IP65, -25 to +45 C, at most 85 dB(A) at 1 m. Droop settable 1-9 %; deadband 0.0-1.0 Hz in 0.05 Hz steps; power factor +/-0.95.

Item	Designation	Rating and particulars
AC skid	4 x ENPCS2520	10.08 MW aggregate.
Step-up transformer	10,100 kVA	0.69 / 33 kV, three-winding, off-load tap changer, LI 170 kV / AC 70 kV, oil-temperature indication, alarm and trip.
MV switchgear	36 kV / 630 A RMU	33 kV rated, 170 kV LI / 70 kV power-frequency withstand, gas tank IP65, enclosure IP3X.
MV station envelope	-	12192 x 2438 x 2896 mm, approx. 49,000 kg.
Controls		
Plant controller	EnOS PPC	Per the filled Volume 2 GTP.
SCADA gateway	To be confirmed by Envision	Addendum No. 01 item 09 requires simultaneous real-time reporting to a minimum of FOUR SCADA master servers - two at the National System Control Centre and two at the Backup NSCC - reportable to any one, any combination or all four, without interrupting existing links and without additional hardware, software, configuration changes or licences. Clarification 27 confirms a gateway is required even where the PPC itself supports IEC 104. Envision to name the gateway product and state its concurrent-session limit and licensing position.

Source: Envision design calculations; filled Volume 2 GTP; 21 August evidence dossier; Addendum No. 01 item 09.

4. Guaranteed performance schedule

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Values are stated at the revenue metering point. Clarification 51 confirms that auxiliary and HVAC loads directly associated with BESS operation are inside the round-trip efficiency calculation, and that general site loads such as CCTV, security and lighting are outside it. Clarification 40 refused a measurement-error tolerance, so no tolerance is claimed below.

Guaranteed parameter	Basis	Guaranteed value
Usable energy at BoL	At the metering point, 0.25P discharge	43.9 MWh installed; 40.0 MWh guaranteed at the Declared Capacity
Discharge duration at rated output	Volume I §3.1(k), at site ambient	4.0 hours at 10 MW at +45 C ambient - Envision to confirm on the thermal case at section 5
Round-trip efficiency, BoL	AC-to-AC, auxiliary-inclusive, monthly	86.9 %
Round-trip efficiency, year 15	AC-to-AC, auxiliary-inclusive, monthly	85.0 % declared; target 85.6 % on the oversized build, Envision to re-rate
State of health, year 15	Against the 68 % MDSC floor	76.7 % on the 10 MW curve; Envision to issue the equivalent curve on the 44 MWh base

Guaranteed parameter	Basis	Guaranteed value
Reactive capability at the PCC	Clarification 24: +/-0.3 x rated active power	+/-3.29 Mvar at power factor +/-0.95, against +/-3.0 Mvar required at a declared 10 MW
Availability	ESA, monthly	97.0 % monthly, guaranteed back-to-back. Envision to supply the reliability block diagram, component failure rates, MTTR and spares holding that support it - see section 11.
Cycle allowance	400 full equivalent cycles per year, floor 20 per month	Cell rated 7300 EFC at 100 % DoD, 9125 at 80 %, 14600 at 50 % (filled GTP)
Self-discharge	Filled Volume 2 GTP	At most 3 % per month
Auxiliary load	At BoL, 4-hour discharge	0.17 MW at 35 C (as calculated); Envision to state the figure at +45 C - this is the open term in the RTE guarantee

Notes: 1. Clarification 5: a monthly RTE shortfall shall not be carried forward or reconciled against later months or the year end. Every month stands alone. 2. Clarification 31: energy lost to frequency or voltage regulation in standby, unscheduled by the grid, is counted in the RTE calculation. 3. Clarification 55(b): energy exchanged providing frequency response, synthetic inertia, oscillation damping and AGC is not excluded from the RTE calculation or from the 400-cycle allowance.

Source: Envision design calculation 5 August 2026; filled Volume 2 GTP; clarification register items 5, 24, 31, 40, 51, 55.

5. Loss stack and the +45 C guaranteed case

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The loss stack below is as stated in the 5 August design calculation for the 11 MW case. The +45 C column is the tender's binding condition and is the single largest open item in this proposal: the received calculation computes auxiliaries at 35 C, while the site envelope runs to +45 C and Volume I §3.1(k) requires four hours at rated output at site ambient.

Loss element	As calculated (35 C basis)	At +45 C site ambient
Calendar degradation, FAT to SAT	97 %	No change expected - Envision to confirm
Usable DC ratio	98 %	Envision to confirm
LV DC cable	99.9 %	Envision to confirm
PCS conversion	98.5 %	Derate expected above 40 C - Envision to state the curve
LV/MV transformer	99.2 %	Envision to confirm at 45 C oil rise
MV cable	99.6 %	Envision to confirm
MV/HV transformer and HV cable	100 % (assumed, BoP scope)	100 % (assumed, BoP scope)
Auxiliary and HVAC	0.17 MW over 4 h at BoL	Envision to state - the RTE guarantee cannot be closed without it

Source: Envision design calculation, 11 MW / 44 MWh, V1.0, 5 August 2026.

6. Thermal substantiation at +45 C - the case Envision must supply

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The system is liquid-cooled, so cell temperature is not ambient temperature. That is the whole of Envision's answer to the cycle-life question, and it is currently unstated. The following is the shape of the case the tender requires.

1. The supplier's own independent test-house bankability study (Issue D Final, 28 January 2026) records approximately 10,000 cycles to 70 % capacity retention at 25 C and 0.25P, but only approximately 4,000 at 45 C. The 20-year / 70 %-retention simulation it endorses is qualified to 25 C, at most 50 % SOC and one cycle per day.
2. The contracted duty is approximately 400 full equivalent cycles per year over 15 years, i.e. roughly 6,000 cycles as a minimum obligation - about 2,000 more than the 45 C figure supports.
3. **Proposed substantiation.** At +45 C ambient the liquid-cooling system maintains a coolant supply temperature of 18 C, holding steady-state cell temperature at or below 30 C at the contracted 0.25P duty, with cell-to-cell spread within 5 C. Cycle life at that cell temperature is therefore materially closer to the 25 C figure than to the 45 C figure, and exceeds the approximately 6,022 EFC the 15-year duty requires. Envision to supply the thermal calculation that establishes this.
4. **Chiller rating point.** The industry convention for this duty is an L45/W18 rating - rated cooling output at 45 C ambient with an 18 C fluid supply. Envision to state the installed chiller capacity at L45/W18 and the electrical draw at that condition, which is the auxiliary figure the RTE guarantee needs.
5. **Contractual backstop.** Where the thermal case cannot be closed before submission, Envision offers a capacity and throughput warranty with an augmentation undertaking triggered on measured capacity falling below the declared curve, at Envision's cost. This converts the exposure into a bankable term and is the correct answer even with more time. Envision to confirm.

7. Grid-forming control - declaration and compliance

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This section is drafted to reconcile two Employer documents that pull in different directions. Annex A A.05.17(i) REQUIRES the BESS to support both grid-following and grid-forming control modes with online switching between them. Volume I §3(c) requires a voltage-sourced, voltage-controlled inverter that must not change to current-controlled operation during normal operation or under any network fault condition. Clarification 12 confirms grid-forming capability is both mandatory and practical for these projects.

1. **Declaration.** The ENPCS2520 supports both grid-forming and grid-following control modes with online switching between them, as Annex A A.05.17(i)

- requires. The plant is delivered, commissioned and operated in GRID-FORM-ING mode as its normal and continuous operating state. It does not change to current-controlled (grid-following) operation during normal operation or under any network fault condition, as Volume I §3(c) requires. Mode transition is available only as a commissioning and maintenance function under NSO's written instruction, and is not an automatic fault response.
2. **Short-circuit ratio.** Grid-forming operation is stable to a minimum SCR of 1.0 and grid-following operation to a minimum SCR of 1.2, as Annex A A.05.17(j) requires. Envision to confirm and to evidence the SCR 1.0 case by EMT simulation - see section 8.
 3. **Reconciliation of existing documents.** The item 61 Grid Compliance List describes GFM and GFL modes with seamless transition at clauses 3.17.4 and 3.17.4.4. That dual-mode capability is REQUIRED by A.05.17(i) and is retained. What is re-pointed is its framing: it is offered as the A.05.17(i) capability, and NOT as the basis for converter robustness under fault, which is delivered by grid-forming operation per Volume I §3(c).
 4. **Document control.** The item 61 Grid Compliance List currently carries the placeholder document number PMD - XXXXXXXX, revision A, and is unsigned, while the checklist requires it signed on company letterhead. Envision to issue a signed, numbered revision.
 5. **Superseded filing.** Checklist item 38 requires grid-forming MODELLING evidence. The document currently filed against it is a black-start technical solution marked 'for reference'. Clarification 37 confirms black start is not mandatory for this project, so that document answers no tender requirement. It is withdrawn from item 38 and replaced by the model package at section 8.

8. Annex A A.05 numeric performance parameters

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Annex A sets hard numeric acceptance criteria. Clarifications 59 and 64 both requested relief on the current ratings - including an explicit request to reduce the 120 % duration from two minutes to one - and both were answered 'Please comply with A.05.02 of Annex A'. No relief is available.

Annex A clause	Requirement	Offered
A. 05.02(a)	PCS AC-side current 110 % continuous, 120 % for at least 2 min, preferably 150 % short-term	110 % continuous, 120 % for 2 minutes, 150 % for 10 seconds, all at up to +45 C. Envision to confirm and state the ambient at which each holds.
A. 05.02(b)	Droop settable 1-9 %; default 4 % where NSO specifies none	Droop settable 1-9 % (filled GTP) - COMPLIES. Default set to 4 %.
A. 05.02(c)-(d)	Synthetic inertia through grid-forming control; inertia time constant no less than 20 s; activation at most 5 ms; flexibly adjustable	Inertia time constant adjustable 2-20 s, factory set at 20 s; inertia response activation under 5 ms, inherent to the voltage-source control and not derived from a frequency or ROCOF measurement. Envision to confirm.

Annex A clause	Requirement	Offered
A. 05.02(e)	Primary frequency regulation response under 0.2 s; active power adjustment deviation at most 2 %	Response under 0.2 s; active power adjustment deviation at most 2 % of Pn. Envision to confirm.
A.05.04	Steady-state 47-52 Hz; extremes 45-55 Hz; 10 s ride-through in the band $47.0 > f \geq 45.0$	Continuous operation 47-52 Hz with ride-through to 45-55 Hz. NOTE: the supplied Sri-Lanka dynamic record carries a 47.5 / 51.5 Hz continuous envelope with 46.9 / 52.1 Hz trip stages, which is NARROWER than A. 05.04 at both ends. Clarification 29 confirms A.05.04 governs. Envision to re-issue the protection settings, or state the equipment limit.
A. 05.17(d)	Autonomous damping of 0.2-2.5 Hz oscillations; active power variation limited to 10-30 % Pn	Autonomous power-oscillation damping across 0.2-2.5 Hz, active power variation limited to 10-30 % Pn, enabled by default. Envision to confirm.
A. 05.17(h)	AGC range -100 % to +100 % Pn, steady-state active deviation at most 2 % Pn; AVC steady-state reactive deviation at most 2 % Pn	AGC -100 % to +100 % Pn with steady-state deviation at most 2 % Pn; AVC steady-state reactive deviation at most 2 % Pn. Envision to confirm.
A.05.17(j)	GFL stable at SCR ≥ 1.2 ; GFM stable at SCR ≥ 1.0	Both met. Evidence by EMT simulation at section 8.
A.05.18	Fault-current contribution bounded by the breaker capability NSO specifies; withstand per A.04	Fault contribution within the 25 kA maximum three-phase fault current at the POC, with withstand for the 550 ms breaker-failure clearing time. Envision to state the contribution magnitude, duration and current-limiting behaviour.

Source: Annex A Functional and Performance Requirement; filled Volume 2 GTP; clarification register items 29, 59, 64.

9. Dynamic simulation models

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Annex A A.05.23(d) states the bid-stage minimum as an ALTERNATIVE: either initial non-site-specific models in both PSS(R)E and PSCAD/EMTDC formats with the tender, OR test results demonstrating V/P/Q response through deep and shallow faults and a +50-degree phase-angle step at SCR 1, 3, 5 and 10 with X/R 5. Failure to meet at least one may result in rejection of the technical proposal. Addendum No. 01 item 14 writes the same either/or into the Volume II Section 5 Compliance Schedule as new item 12. Envision holds both models, so route (I) is satisfied at bid stage.

Deliverable	Status	Particulars
PSS(R)E RMS model	HELD - submitted	ENVSG01_20260327_PSSE_V35.dll (grid-forming PCS), ENPPC_260415_PSSE_V35.dll (PPC), and the Sri-Lanka dynamic record ENVSG_PPC_2520_260416_LKA.dyr instantiating ENGFM01 (2520 kW base) and BNPPC_GFMV3. Manual V1.4a, 11 August 2026. Runs in PSS(R)E 35.x as A.05.23(c) requires.

Deliverable	Status	Particulars
PSCAD/EMTDC EMT model	HELD - but grid-FOLLOWING variant	PCS2520x4_UPPC_x64_260605aBB.pscx (4 x ENPCS2520 at 690 V / 50 Hz on one 10.1 MVA skid), PCSControllerInterface.dll, five .obj and one .lib interface objects. Requires PSCAD V5.0 with Intel Fortran XE 15+, 1-200 us timestep, 50 us recommended - meets A.05.23(c).
Grid-forming EMT model	REQUIRED - not held	The EMT model supplied is titled for the GFL PCS variant, its model block is labelled GFL-PCS, and the converter is represented as a CURRENT SOURCE. A current-source model cannot demonstrate voltage-source behaviour under fault, and the PSS(R)E grid-forming manual itself records that EMT is the better tool for fault ride-through. Envision to confirm whether a grid-forming PSCAD model of the ENPCS2520 exists and to issue it. If none exists, that answer is needed immediately.
SCR sweep and phase step	Post-award deliverable	V/P/Q traces through deep and shallow faults and a +50-degree phase-angle step at SCR 1, 3, 5 and 10 with X/R 5, plus the A.05.17(j) grid-forming case at SCR 1.0. Not required at bid stage where both models are submitted. Due within one month of ESA execution - 14 January 2027 - under Addendum No. 01 item 12.
PowerFactory models	NOT required	A.05.23(b) names a third tool, but clarification 35 confirms: 'It is compulsory to provide PSSE and PSCAD models.' PowerFactory is therefore outside the compulsory set.

Notes: 1. Addendum No. 01 item 06 adds failure to submit the Dynamic Model Test results, in a form and level of detail acceptable to NSO, within one month of ESA execution to the grounds on which the Performance Security may be forfeited. The grid-forming EMT gap is therefore a security-backed post-award obligation on a one-month clock, not merely an evidence gap. 2. Document hygiene noted for correction: the PSCAD manual describes the compiled DLL using wind-turbine source-code wording, and section 4 of the PSS(R)E manual renders as a Word undefined-bookmark error in the released PDF.

Source: Annex A A.05.23; Addendum No. 01 items 06, 12 and 14; clarification 35; 21 August evidence dossier.

10. Fire safety and thermal runaway

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Element	Status	Particulars
UL 9540A cell level	HELD	Cell-level report to UL 9540A-2019.
UL 9540A module level	HELD	Module-level report to UL 9540A-2026, 54 pp, issued 25 June 2026. Recorded results: peak heat release rate 46.28 kW; peak smoke release rate 0.5457 m ² /s; total smoke release 29.98 m ² ; total hydrocarbons 432.6 L; module weight loss 1.8 kg; no flaming observed. Cell-to-cell thermal runaway and propagation occurred and was contained by the module design; cell vent gas was determined flammable at cell level.
Next test level	REQUIRED - not held	The module report's own conclusion states that a further test level is required. Under the sixth edition of UL 9540A, published 13 March 2026, the unit-level test is no longer required for non-residential BESS, and the installation-level large-scale fire test at Section 10 is the integrated fourth evaluation level, aligned to NFPA 855 and intended to demonstrate that fire does not propagate between ESS units. Envision offers a dated commitment to the Section 10 installation-level test, with the module-level result as the interim compliance basis. Physical burn testing cannot complete before 4 September. Envision to confirm the route and the date.
Fire Protection System Specification	HELD - scope mismatch	V2.0, 21 May 2026. The header scopes ENS-D10, while the offered configuration uses ENS-D06G and ENS-D10G containers.

Element	Status	Particulars
D06G coverage	REQUIRED	Envision to re-issue the specification with a header scoping both offered container types, or to confirm coverage of ENS-D06G in writing.

Source: 21 August evidence dossier, checklist item 45; UL 9540A sixth edition (13 March 2026).

11. Standards and certification register

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Sixteen certificates and reports are held across checklist sections C and D - five DC-side and eleven AC-side. Clarification 62 opens the route for the remainder: where the exact proposed package is newly introduced and certification is still in progress, the bidder may submit currently valid certifications for the applicable cells, racks, major components or established product family, TOGETHER WITH documentary evidence of the relationship and technical equivalence to the offered package. That is an evidence route, not a waiver.

Standard	Declared position	Action
IEC 62619 (CB scheme)	Certified, cell level	Held. Certificate is in the cell-manufacturing entity's name.
UL 1973	Component recognition, cell level (BBGA2)	The certificate states that UL Recognized components are incomplete in certain constructional features. It is not system certification and must not be presented as such. Envision to supply the equivalence evidence bridging cell-level recognition to the offered system under clarification 62.
UL 9540 (system level)	Partially comply / future	Envision to state a realistic certification date and the interim equivalence basis.
IEC 62620, IEC 62933-5-2	Not certified - 'Will finish before 2026'	STALE: the date has passed. Envision to re-date with realistic completion dates - a stale commitment reads worse than an honest revised one.
IEC 62902	Not comply - ISO 7010 labelling offered instead	Envision to state the equivalence argument.
IEC 62485-5	Not comply - IEC 62619 / 61000-6-2/-6-4 / 62620 offered	Envision to state the equivalence argument.
IEC 62933-1	Comply, no certificate - 'do not plan to certificate'	Envision to supply a signed declaration of compliance in lieu.
IEC 62933-2-1	Not comply - framework standard	Envision to state the equivalence argument.
IEEE 1547-2018	Partially - only US-version products tested	Envision to state which provisions are met by the offered product.

Standard	Declared position	Action
IEEE 2800-2022	Declined - 'US region-specific, not required for other markets'	This is the substantive refusal. IEEE 2800 is the transmission-level IBR interconnection standard whose grid-forming provisions are the natural reference for this tender's mandatory grid-forming requirement, and declining it on market-scope grounds will be read closely. Envision to state the equivalence argument clause by clause against EN 50549-2 and G99. NOTE: no published equivalence mapping between these standards exists, so the argument must be constructed, not cited. IEEE has issued amendment IEEE 2800a specifically to reduce barriers for inverter-based resources with grid-forming equipment; it is the right reference to engage.
UL 1741-SB	Declined - US-only; EN 50549-2 / G99 offered	Envision to state the clause-level equivalence argument.
IEC TS 62786-3	Not applicable - string-PCS standard; centralised PCS offered	Position is sustainable as stated.
Sri Lanka Grid Connection Code (July 2024)	Checklist item 36 marked Not Received, no remark	STALE STATUS: item 61, the Grid Compliance List of 11 August 2026, post-dates the checklist and largely answers it. Bidder to update the workbook.

Source: 21 August evidence dossier, checklist sections C and D; clarification 62.

12. Capacity maintenance, availability and end of life

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1. Capacity Maintenance Plan (Volume I §3.1(m)). Envision to issue a plan, not a curve. It must state beginning-of-life energy at the Termination Point against the contracted rating; the augmentation schedule in MWh and timing, or an express and substantiated statement that none is required; module and rack replacement intervals; the augmentation and outage interface with NSO; and end-of-life decommissioning and recycling commitments.
2. Augmentation. Checklist items 55 and 56 declare that no augmentation and no replacement are needed during the lifetime, and both status cells are blank. That declaration is supportable only near 25 C cell temperature. Envision to substantiate it against the thermal case at section 5, or to replace it with a trigger-based augmentation undertaking with reserved space, compatible future cell supply, outage durations and spares.
3. Availability. Envision to supply a reliability block diagram, component failure rates, redundancy, MTTR, planned-maintenance schedule and spares holding sufficient to support 97 % monthly availability. This carries unusual weight: clarification 54 confirms there is no aggregate cap on liquidated damages per Contract Year or over the 15-year Term, and that capacity-charge deductions for missing 97 % availability are NOT liquidated damages and fall outside the monthly cap - so the capacity charge for a month may be reduced to LKR 0. The operator reference record shows PCS as the dominant

availability risk: 68-70 hours of PCS downtime against 0 hours of cell-fault downtime.

4. **End of life.** The document currently filed at checklist item 57 is a mechanical disassembly work instruction; a keyword sweep of it returns zero occurrences of recycling, disposal, waste, take-back, second-life or Sri Lanka environmental regulation. Envision to supply a genuine commitment: the take-back or recycling route, the receiving facility and its jurisdiction, treatment of the LFP chemistry, and express reference to the applicable Sri Lankan environmental regulations.
5. **Service life (Volume I §3.1(l)).** Envision to confirm 20-year design life for non-battery equipment and 15-year for cells, modules and racks under the ESA duty cycle. The 20-year letter currently held is self-certified - the checklist itself records 'No certification to prove this but we provided a statement'.

13. Qualification and corporate

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Requirement	Position	Action
Manufacturer track record (Volume I clause 2.7.3)	Envision Energy letter: 18.7 GW / 52.6 GWh contracted; 33.5 GWh shipped; 20.65 GWh at COD	Clarification 58(a)-(b) settles that the threshold is CUMULATIVE GLOBAL INSTALLED / COMMISSIONED volume, 'because the requirement says installed'. The qualifying figure is therefore the 20.65 GWh at COD - not the 52.6 GWh contracted or 33.5 GWh shipped. The bid should quote 20.65 GWh.
Whose experience qualifies	Track record in Envision Energy's name; EOI signed by an affiliated supply entity; cell certificates in the cell-manufacturing entity's name	Clarification 58(e): 'The qualification should be satisfied based on the manufacturer/component supplier's experience, and therefore need not be experience of the Project Proponent itself.' Residual: which group entity holds it. Envision to confirm in writing that Envision Energy's track record is available to and may be relied upon by the entity that signed the EOI letters, with any parent guarantee needed to make that reliance effective.
Manufacturer's Authorization Letter	Checklist item 58 marked Received; three EOI letters present; NO MAL in the package	Envision to issue the MAL on company letterhead.
Offered PCS track record	22 PCS reference projects listed, using ENPCS 2750, 3450, 3300 and 2500. The offered ENPCS2520 appears in none	Envision to explain the ENPCS2520's relationship to the referenced family and identify any commissioned ENPCS2520 units with project, location and commissioning date. Clarification 58(c) accepts a manufacturer declaration on letterhead carrying product, installed quantity, project, location, commissioning date and manufacturer.
Reference contact details	Client contact column empty for every PCS and PPC row; workbook self-labelled '(part)'	Envision to complete the contact column and issue the workbook in full.

Requirement	Position	Action
Operating references	Four operator letters. Two named-grid projects report 98.2 % availability to 1 March 2024, with 0 h cell-fault, 36-43 h DC-system, 68-70 h PCS and 1 h EMS downtime	None of the four demonstrates grid-forming operation. Envision to identify any grid-forming reference installation.

Source: 21 August evidence dossier; clarification register item 58.

14. Commercial framework required from Envision

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The Project Company's exposure under this ESA is unusually open, and the ESA cannot be amended after the Letter of Award (Volume I §7.1, confirmed by clarification 30). Whatever back-to-back support Envision provides must be agreed before submission.

ESA exposure	Established by	Back-to-back support required
No aggregate cap on liquidated damages, per Contract Year or over the 15-year Term	Clarification 54	Envision to confirm an aggregate liability position adequate for performance-caused losses, without excluding model or control failures.
Capacity charge may be reduced to LKR 0 in a month missing 97 % availability, outside the monthly LD cap	Clarification 54	Back-to-back availability warranty with response times, spares, remote diagnostics and OEM-caused LD indemnity.
RTE liquidated damages at 150 % of the peak-time 33 kV GP tariff on excess losses, assessed monthly with no reconciliation	Volume I §2.8(iii); clarifications 5 and 40	RTE guarantee at the contractual meter, correction plan, test protocol and back-to-back loss compensation. No measurement tolerance is available.
No termination compensation or buy-out for NSO default, political force majeure or prolonged natural force majeure	Clarification 52	Envision to confirm supply and warranty continuity terms that survive a termination in which the Project Company recovers nothing.
Flat LKR capacity charge for 15 years with a single 85 % USD adjustment at ESA signing; P2 fixed and not moved to LC settlement	Clarification 16	Price validity through award and ESA signature, defined FX basis, capped LTSA and spares escalation, transparent local/foreign split.
No CCR adjustment or penalty-free withdrawal for tax or regulatory change after the Closing Date	Clarification 19	Envision to hold pricing on that basis or state the assumptions it is conditioned on.
Supplier substitution after award refused	Clarification 73(b)	Envision to confirm supply-chain security for the exact nominated equipment for the full delivery window.

Source: NSO RFP Volumes I and III; clarification register items 5, 16, 19, 30, 40, 52, 54, 73.

15. Bidder-side actions - not Envision's scope

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Recorded here so nothing is lost between the two workstreams. These are the Project Proponent's, not Envision's.

1. **GRID INTERCONNECTION CONFIRMATION LETTER** - most urgent. Addendum No. 01 items 02 and 08 make it mandatory where Option 2 is selected under Volume II Section 11, and a MAJOR DEVIATION under Volume I clause 6.3.1 if absent. Item 07 required it to be requested from the Provincial Director of EDL at least 21 days before closing - i.e. by 14 August 2026 - and EDL is not liable for delays. Confirm immediately which Option is taken and whether the letter was requested.
2. **PCA3 registration**. Addendum No. 01 item 10 adds failure to submit the Certification of Registration under the Public Contract Act No. 3 of 1987 to the grounds for disqualification.
3. **Submission arithmetic**. A separate RFP purchase per proposal (clarification 46); each proposal a complete and separate submission (clarification 70); LKR 600 million financial resources per project, cumulative - LKR 1,200 million for two (clarification 69); Proposal Security addressed to the CEO of NSO (clarification 41) and valid to 3 March 2027 (Addendum item 03).
4. **Joint-venture structure**. Clarification 60: any technical or manufacturing partner that is a FORMAL JV member must meet 25 % of the LKR 600 million regardless of its role. To avoid that, it must be structured as a manufacturer, supplier, technical partner or subcontractor, not a JV member. FIN-1 to FIN-4 are completed by each formal member.
5. **Shared interconnection**. Clarifications 20 and 75 contradict each other - 20 permits a common grid interconnection line for two projects at one GSS, 75 says only one BESS project can connect to a single 33 kV feeder. The window to ask closed on 25 August. Cost any two-project bid on the conservative reading (separate feeders) and state the assumption in the bid.
6. **Capacity Charge Rate unit**. Clarification 65 answers 'LKR/MW/month' and states the proposed rate as Y LKR/MW/month, then writes the applicable-rate formula $0.15xY + (0.85xYxP2/P1)$ in LKR/MWh/month - a different unit in the same answer. Quote in LKR/MW/month and complete the SSCL and VAT-18 % rows added by Addendum item 13.
7. **Tracking workbook**. Correct the Summary COUNTIF range from F2:F65 to F2:F69 (it reports 38 Received against a true 41), restore missing items 50-53, set statuses for items 55-57, and update item 36 against item 61.

16. Basis of statement - what is sourced and what is drafted

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This section exists so that no reader has to infer the status of any statement in this draft from its tone.

Colour	Meaning	How it may be used
Black	Sourced. Traceable to a document held in the corpus with its SHA-256 recorded in MANIFEST.sha256 - an NSO tender volume, Addendum No. 01, Annex A, the clarification register, an Envision design calculation, the filled Volume 2 GTP, or the 21 August evidence dossier.	May be relied upon subject to the source-governs caveat on every page.
Red	DRAFTED GAP-FILL. Not found in any received Envision document. Written the way the tender requires it to read, so Envision can confirm, correct or replace it.	MUST NOT be submitted, quoted to NSO, or relied upon until Envision has adopted it in writing. Red values are drafting placeholders that reflect what the tender demands - they are NOT representations about the offered product and were not supplied by Envision.

17. Source register

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Source	Role	SHA-256 (first 12)
NSO RFP Volume I	Controlling tender volume	fb61a4f827a0
NSO RFP Volume II	Controlling tender volume	41644c207213
NSO RFP Volume III (Model ESA)	Controlling tender volume	d3413e9d1a1b
Addendum No. 01, 7 August 2026	Controlling amendment - overrides what it amends	2b99b1e507d8
Annex A Functional and Performance Requirement	Controlling technical annex	be599073f597
Clarification register, 76 items, 21 August 2026	Controlling clarifications (image-only scan; read from page images)	73fbaca1a7e7
Envision design calculation 10 MW / 40 MWh, 29 July 2026	OEM evidence - superseded 5 August	7281d964654e
Envision design calculation 11 MW / 44 MWh, 5 August 2026	OEM evidence - the offered configuration	0cf77ec5d761
Envision functional-requirements checklist	Supplier declaration - not evidence	8236806c21f6
Bidder evidence dossier, 21 August 2026	OEM evidence - held outside the repository, manifest only	not committed

Source: docs/source_materials/nso_bess_250mw_2026/MANIFEST.sha256

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PDF variant pdf/ua-1; house stylesheet applied: True.
Font substitution: none; house fonts embedded: UNVERIFIED (poppler unavailable).
Assembled only from the DutchBay NSO corpus. Red text is drafted gap-fill, not supplied by Envision, and is not a representation about the offered product.